

Department of Computer Science

TDT4259 - Applied Data Science

Aneo
Electricity Consumption Forecasting for a Retailer

Authors:
Luca Agostini
Riccardo Lori
Beatrice Losi
Manuel Maiuolo
Marie Schwabe
Henriette Strømsvåg

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1 Problem definition and business understanding

1.1 Context

'The power market ensures that resources are used efficiently, that security of supply is maintained, and helps keep electricity from becoming more expensive than necessary' (Energidepartementet 2026, own translation).

The Nordic wholesale market consists of three sequential markets: the day-ahead market, the intraday market and the balancing markets. The day-ahead market, operated by Nord Pool, is where most of the volume is traded. Between 08:00 and 12:00, participants submit buy and sell bids for each quarter-hour of the following day, and prices are set where supply meets demand, subject to available transmission capacity. After the auction, participants can adjust their positions in the intraday market until one hour before delivery. Remaining deviations are handled by Statnett, which as system operator uses the balancing markets to keep production and consumption in balance in real time. (Energidepartementet 2026)

1.2 Problem and opportunity

For an electricity retailer, the market design means that customers' consumption must be bought before it is known. Any difference between the purchased volume and actual consumption becomes an imbalance, settled at an imbalance price that is volatile and hard to predict. Accurate consumption forecasts are therefore essential for the retailer to operate cost-effectively in the market.

Forecasting this consumption is challenging. Consumption varies between price areas and with season and temperature, as well as with the day of the week and the time of day, for example with higher consumption in the morning when people get ready for work. In addition, metered consumption data is only available with a delay of about five days, so the most recent consumption trends cannot be used when forecasting the next day's consumption. Further constraints are discussed in Section 1.5.

In this project, we take the role of Aneo's data science team. Aneo delivers consumption forecasts to an electricity retailer under a fixed-price contract. The task is to forecast the retailer's hourly consumption for each hour of the next day in six price areas (NO1–NO5 and FI). The forecast must be delivered around 10:00, so that the retailer can use it in its day-ahead bids before the auction closes at 12:00. From a data science perspective, this is a regression problem consisting of predicting hourly consumption for the next day, using only information available at 10:00 on the day of prediction. As a baseline, we assume consumption to be equal to the same hour one week earlier.

1.3 Stakeholders and decision

Three stakeholders are considered: Aneo as the forecast provider, the retailer as the end user of the forecast, and indirectly the retailer's customers. The retailer makes the final decision on how much electricity to buy in the day-ahead market, and the forecast serves as a decision-support tool rather than an automated decision. Because Aneo operates under a fixed-price contract, it does not bear the imbalance costs directly. The value of an accurate forecast for Aneo is therefore indirect: lower imbalance costs for the retailer increase customer satisfaction and the likelihood of contract renewal. The key business question is therefore not only which model predicts consumption most accurately, but whether the forecasting solution provides sufficiently reliable and valuable forecasts for the retailer to continue using the service.

1.4 Success criteria

Before developing a model, success criteria must be defined in order to assess its value. The criteria are divided into four tiers: forecast quality, business value, operational reliability and customer relationship.

- **Tier 1: Forecast quality**

These metrics measure how much the forecast deviates from actual consumption. All metrics are evaluated on a chronological test period.

Let y_t denote actual consumption and $\hat{y}_t$ the forecast in hour t, with n hours in the evaluation period:

$$\text{MAE} = \frac{1}{n}\sum_{t=1}^{n}|y_t - \hat{y}_t|, \qquad \text{RMSE} = \sqrt{\frac{1}{n}\sum_{t=1}^{n}(y_t - \hat{y}_t)^2}, \tag{1}$$

$$\text{WAPE} = \frac{\sum_t |y_t - \hat{y}_t|}{\sum_t y_t}, \qquad \text{NMBE} = \frac{\sum_t (\hat{y}_t - y_t)}{\sum_t y_t}. \tag{2}$$

Table 1: Forecast accuracy metrics (Tier 1).

Metric	**What it measures**	**Relevance in this case**
MAE	Average absolute error (MWh)	Primary metric. With symmetric loss, it directly represents the average imbalance per hour.
RMSE	Error with squared deviations (MWh)	Penalises large errors more heavily, revealing occasional severe misses.
WAPE	Absolute error relative to total consumption (%)	Makes performance comparable across price areas with different consumption levels.
NMBE	Signed error relative to total consumption (%)	Reveals systematic over- (> 0) or underforecasting (< 0); target close to zero.

All metrics are also computed for the baseline (same hour one week earlier), and the model is considered successful only if it outperforms the baseline both overall and in each price area. In addition to aggregate values, MAE is reported by hour of day, by price area and for high consumption periods, to reveal whether errors are concentrated when imbalances are most costly.

Forecast accuracy is, however, only a proxy for business value, which is addressed in Tier 2.

- **Tier 2: Business and financial value**

  Although Aneo operates under a fixed-price contract, reducing the retailer's imbalance costs increases customer satisfaction and the likelihood of contract renewal (see Section 1.3). Business value is therefore assessed with the following KPIs:

  - *Imbalance volume reduction:* the reduction in total absolute imbalance, both in MWh and as a percentage, compared with the baseline.
  - *P95 imbalance:* the absolute error that is exceeded in only 5% of hours. While MAE describes the typical error, P95 describes the size of the errors in the worst hours, indicating how much imbalance the retailer should be prepared to handle. As a rule of thumb, P95 is approximately 2.5· MAE for normally distributed errors; a higher ratio indicates more frequent extreme errors than expected.
  - *Share of forecasts within an acceptable range:* the share of hours where the error is within ±5% of actual consumption.
  - *Reliability across price areas and periods:* the KPIs above should improve on the baseline in all price areas and seasons, not only on average (see Tier 1).
  - *Simulated imbalance cost savings:* the imbalance cost avoided (NOK) compared with the baseline. Since the retailer pays the spot price for its purchased volume regardless, the cost of an imbalance is approximated by the difference between the imbalance price and the spot price. This places more weight on errors in hours where imbalances are most costly. For each price area a and hour t, the simulated imbalance cost of a forecast is

    $$C = \sum_a \sum_t |\hat{y}_{a,t} - y_{a,t}| \cdot |p^{\text{imb}}_{a,t} - p^{\text{spot}}_{a,t}|, \tag{3}$$

    where $p^{\text{imb}}_{a,t}$ and $p^{\text{spot}}_{a,t}$ are the imbalance and spot prices (NOK/MWh). The simulated cost savings are

    $$\Delta C = C_{\text{baseline}} - C_{\text{model}}. \tag{4}$$

This approximation assumes that over and underforecasting cost the same per MWh, that the retailer does not adjust its position in the intraday market, and that the retailer's imbalance does not affect market prices. As imbalance prices are not available (see Section 1.5), the price difference is approximated as $|p^{\text{imb}}_{a,t} - p^{\text{spot}}_{a,t}| \approx k \cdot p^{\text{spot}}_{a,t}$, where k is an assumed share. As no reliable estimate of the imbalance price premium is available, $k = 0.2$ is used as a base case, with a sensitivity analysis for $k = 0.1$ and $k = 0.5$. Since k scales the cost of both the model and the baseline equally, it affects the size of the savings but not the ranking of the forecasts.

- *Value relative to contract price:* The contract fee is not given in the case. Instead, the simulated savings are used to indicate the maximum fee the retailer would be willing to pay for the service. For the service to be viable, this must exceed both a reasonable market price for forecasting services and Aneo's own cost of developing and operating the model.

- **Tier 3: Operational reliability**

  The forecast must be delivered by 10:00 local time every day, including weekends and holidays, with a target of 100% on-time delivery. If the model fails or input data is missing, the baseline forecast is delivered automatically, since a simple forecast is better than none. Monitoring and retraining are described in Section 6.

- **Tier 4: Customer relationship**

  In the longer term, the success of the service is reflected in retailer satisfaction and customer retention. These cannot be calculated directly from the dataset, but they follow from the KPIs above: a more accurate forecast leads to smaller imbalances, lower financial exposure for the retailer and a higher perceived value of the service, which in turn increases the likelihood of contract renewal. Since the retailer combines the forecast with its own market assessments, the forecast should also be interpretable, for example by showing the main drivers such as temperature and calendar effects. Together with reliable delivery (Tier 3) and stable performance across price areas, this builds the retailer's trust in, and continued use of, the forecasts.

1.5 Constraints and scope

The success criteria above define what the forecasting solution should achieve. Achieving them is, however, subject to several practical, temporal and business constraints, which also define the scope of the project.

- **Forecast horizon and information:** The forecast is made at 10:00 for 14–37 hours ahead, using only information available at that time. Temperature inputs must therefore be weather forecasts, not observed values.

- **Delayed consumption data:** Metered consumption is available with a delay of about five days, so the most recent consumption lies at least six days before the target day.

- **Data scope:** The data covers one retailer's customer portfolio, not total consumption in each price area. Consumption may shift as the retailer gains or loses customers, and results may not generalise to other retailers.

- **Simplifications:** Forecasts are made at hourly resolution, although the day-ahead market has used 15-minute resolution since 2025 (Energidepartementet 2026). Historical spot prices are available, but imbalance prices are not. The imbalance cost in Equation (3) is therefore approximated by assuming that the imbalance price deviates from the spot price by a fixed share k of the spot price.

- **Out of scope:** The project forecasts consumption only. Price forecasting is excluded, as prices are set by the market as a whole and are not part of the service Aneo delivers. The retailer's bidding strategy is also excluded, since the retailer makes the final purchasing decision and may combine the forecast with its own market assessments. Finally, Aneo's production planning is outside the scope, as the forecast is delivered as a service to the retailer, not for Aneo's internal operations.

2 Data understanding and background

2.1 Domain background and structural dynamics

Electricity consumption in the Nordic wholesale power market is driven by physical climate conditions, national heating infrastructures, and macroeconomic shocks (Energidepartementet 2026). Understanding these mechanisms is crucial to design effective features and avoid model misspecification across price areas.

Space heating infrastructure disparity. In Norway, electricity demand is exceptionally weather-sensitive. Driven by historically abundant hydropower, direct electric resistance heaters and heat pumps account for the vast majority of residential and commercial space heating (Statistisk sentralbyrå 2023a). Consequently, winter sub-zero temperatures trigger dramatic load surges across Norwegian bidding zones (NO1–NO5). Conversely, urban Finland (FI / Helsingfors) relies predominantly on municipal district heating networks (*kaukolämpö*), which supply over 90% of apartment buildings with thermal energy from combined heat and power plants (Finnish Energy 2022). As a result, electricity consumption in Helsingfors is largely decoupled from ambient temperature, exhibiting a much flatter annual profile and minimal cold-weather elasticity.

Macroeconomic context and activity cycles. The dataset period (April 2022 to April 2023) coincided with the acute phase of the European energy crisis following the Russian invasion of Ukraine and subsequent gas supply disruptions (International Energy Agency 2022). Elevated continental electricity and fuel prices spilled into interconnected southern Norwegian bidding zones (NO1, NO2, NO5), inducing voluntary demand conservation and behavioral load adjustments. Simultaneously, regular human routines produce deterministic intra-day diurnal curves: a morning ramp (06:00–09:00 UTC), an evening peak (17:00–20:00 UTC), and a night trough (01:00–04:00 UTC). While activity drops slightly on off-days (weekends and public holidays), empirical analysis reveals that seasonal temperature swings vastly dominate calendar-day variance.

2.2 Data sources and descriptive statistics

The primary dataset provided for this project comprises hourly records from April 2022 to April 2023 across six Nordic bidding zones: NO1 (Oslo), NO2 (Stavanger), NO3 (Trondheim), NO4 (Tromsø), and NO5 (Bergen) in Norway, plus FI (Helsingfors) in Finland. It contains exclusively two variables: **metered consumption (y_t, MW)** and **exogenous temperature forecasts (T_t, °C)**. The first one represents the realized hourly electricity consumption for the specific retailer's customer portfolio. The second one is the hourly numerical 2-meter air temperature forecast assigned to each bidding area.

To evaluate the business and financial impact defined in Tier 2 (simulating avoided costs in Equation 3), we independently augmented the dataset by collecting historical hourly day-ahead spot prices (p_t^{spot}) from the ENTSO-E Transparency Platform (ENTSO-E 2023), converted to NOK using official exchange rates (Norges Bank 2023).

Table 2 reports descriptive statistics across bidding areas. Portfolio volume varies substantially: Oslo accounts for the highest load (mean 6.88 MW, peaking at 18.05 MW), whereas Tromsø is the smallest (mean 1.34 MW). Stavanger's demand exceeds Trondheim's, reflecting the retailer's regional market share rather than city population, as Trondheim is substantially more populous than Stavanger ($\approx 212,000$ vs. $\approx 145,000$ inhabitants in 2023, (Statistisk sentralbyrå 2023b)).

Table 2: Descriptive statistics of hourly consumption (MW) and temperature (°C) across bidding zones.

			Consumption (MW)			**Temperature (°C)**		
Zone / City	**Period**	**Obs.**	**Mean ± Std**	**Min**	**Max**	**Mean ± Std**	**Min**	**Max**
NO1 / Oslo	07/04/22–02/04/23	8,641	6.88 ± 4.67	1.52	18.05	7.39 ± 8.77	-15.4	29.8
NO2 / Stavanger	07/04/22–02/04/23	8,641	3.30 ± 2.11	0.83	8.76	8.55 ± 5.84	-9.2	27.0
NO3 / Trondheim	07/04/22–02/04/23	8,641	2.28 ± 1.14	0.62	5.10	6.21 ± 7.62	-17.3	30.5
NO4 / Tromsø	07/04/22–02/04/23	8,641	1.34 ± 0.56	0.37	2.55	3.51 ± 7.97	-14.6	29.8
NO5 / Bergen	07/04/22–02/04/23	8,641	1.88 ± 1.16	0.44	4.65	7.75 ± 6.56	-15.6	29.3
FI / Helsingfors	14/07/22–02/04/23	6,289	6.08 ± 0.92	0.00	7.86	5.66 ± 7.59	-10.1	24.0

2.3 Exploratory data analysis and important variables

Scale disparities and seasonal trajectory. Figure 1 highlights the wide volume variation between bidding areas and the pronounced U-shaped seasonal curve across Norway, where winter demand surges two- to threefold compared to summer baselines. Additive time-series decomposition confirms that macro seasonal dynamics govern this trajectory, accompanied by consistent 24-hour diurnal load cycles. Residual variance remains small in summer but expands significantly during winter cold spells.

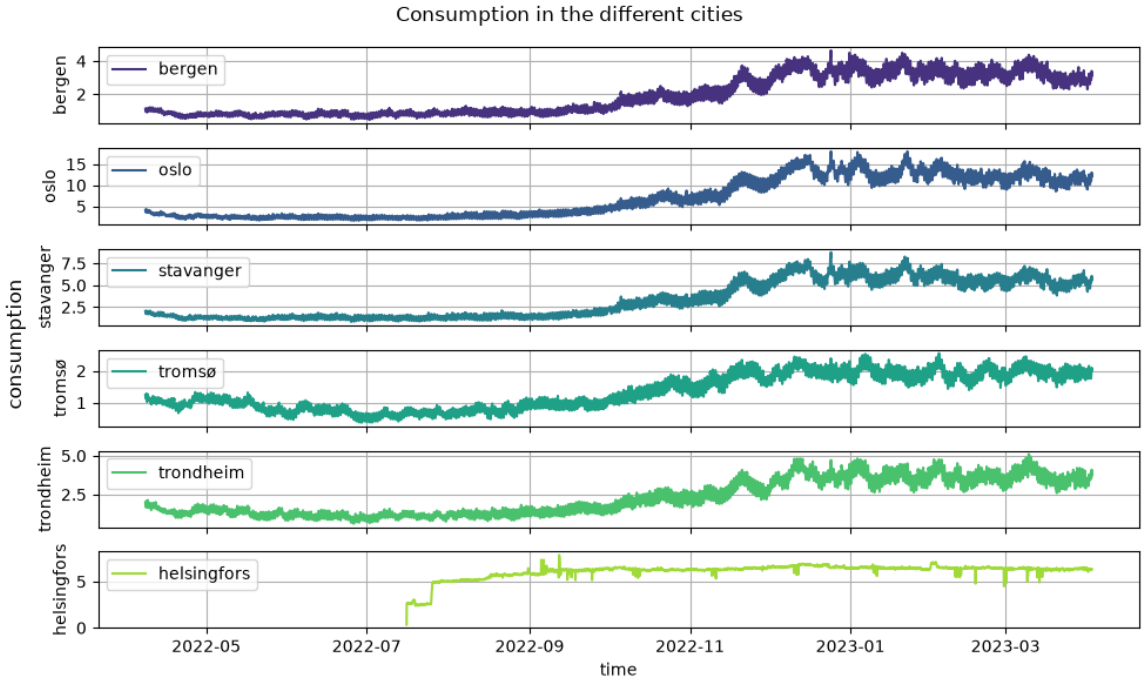


Figure 1: Hourly electricity consumption series across bidding zones (April 2022 – April 2023).

Non-linear heating curves and Finnish divergence. Figure 2 presents the city-by-city scatter plots of consumption against temperature. This reveals a clear non-linear 'heating curve': above 15 °C, electricity demand remains flat at baseload levels, confirming negligible air conditioning in the Nordics. Below 15 °C, demand rises linearly, accelerating sharply below 0 °C as electric heating operates near peak capacity. Crucially, the city-by-city breakdown illustrates that Helsingfors is completely insensitive to sub-zero temperatures (narrow IQR = 0.28 MW), empirically validating the dampening role of municipal district heating.

Important variables and feature drivers. Guided by exploratory analysis and operational constraints, the predictive feature set relies on four key groups:

- *Target variable:* Hourly metered consumption (y_t, MW).
- *Weather predictors and derivatives:* Exogenous temperature forecast (T_t), daily mean temperature ($\bar{T}_{\text{day}}$), and 24-hour thermal difference ($T_t - T_{t-24}$). Non-linear transformations (T^2, heating-degree thresholds below 15 °C) are critical to model the physical heating inflection.
- *Calendar and temporal features:* Hour of day (capturing the morning 06:00–09:00 and evening 17:00–20:00 peaks), day of week, and annual seasonality. While weekday and off-day load distributions almost completely overlap, weekly patterns are captured naturally by autoregressive lags.
- *Lagged consumption features ($\geq$ 144 h):* Autoregressive lags respecting the 5-day latency constraint (lag_{144}, lag_{168}, lag_{192}, lag_{336}) and 7-day rolling statistics ($roll_mean_{7d}$, $roll_std_{7d}$ shifted by 144 hours) to establish portfolio baselines.

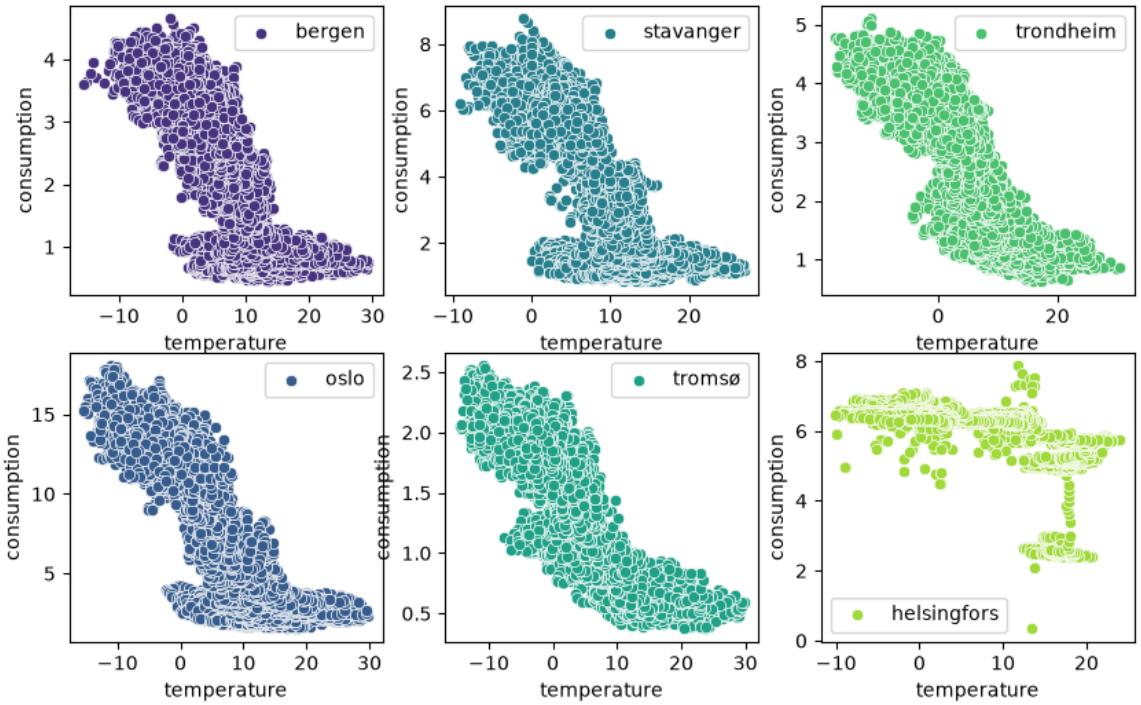


Figure 2: Consumption versus temperature by bidding zone, illustrating the 15 °C inflection and Finnish thermal invariance.

2.4 Data quality issues, anomalies and remediation

Rigorous screening was performed to verify data integrity and isolate actionable quality issues:

- **Limited temporal sample:** The dataset spans only 12 months, observing only a single winter cycle and precluding multi-year seasonal cross-validation.
- **Asymmetric temporal coverage:** Norwegian zones contain 8,641 consecutive hours (April 7, 2022 to April 2, 2023), whereas Helsingfors contains only 6,289 records, commencing three months later on July 14, 2022 due to subsequent customer onboarding.
- **Absence of internal missing records:** Once telemetry was active, zero missing timestamps or NaN values occur across all six locations (freq = 'h').
- **Physical extremes vs. spurious outliers:** Statistical screening ($1.5 \cdot \text{IQR}$ and 3σ) flagged Oslo's winter peaks ($\sim$ 18 MW, on Christmas Eve and -12 °C freeze) as global outliers. However, domain validation confirms these represent authentic space heating demand. Removing or winsorizing them would dangerously understate balancing risks in Tier 2 financial metrics. Likewise, thermal extremes (-17.3 °C in Trondheim, +30.5 °C in summer) are verified records essential for training weather elasticity.
- **Telemetry initialization artifact in Helsingfors:** Exactly 32 consecutive observations report consumption = 0.0 MW in Helsingfors between 2022-07-14 21:00 and 2022-07-16 04:00 UTC at the very beginning of the series, reflecting telemetry initialization latency.
  Remediation: Rather than imputing synthetic values or introducing time-series gaps, we trim the Helsingfors start date to **2022-07-16 05:00:00 UTC**, ensuring complete, unbroken hourly records across all areas.
- **Weather forecast opacity:** Temperature values are provided as forecasts, but metadata regarding the exact numerical model run time and generation latency is unobserved.

2.5 Assumptions and operational limitations

The forecasting task operates under several binding assumptions and practical limitations:

- **Operational 5-day metered latency:** At 10:00 on day D, the retailer bids for the 24 hours of day $D+1$ (lead times 14 to 37 hours ahead). Because metered consumption requires approximately five days of validation, the latest verified load available originates at $t-144$ (day $D-5$). This fundamental operational constraint eliminates standard short-term autoregression (lag_{24}, lag_{48}) and immediate error feedback, forcing models to rely heavily on weekly seasonality (lag_{168}) and weather forecasts.
- **Single-year horizon and seasonal generalizability:** Spanning only one annual cycle, the dataset captures a single winter. Models cannot assess performance across varying winter severities (mild vs. severe winters) or decouple 2022 energy crisis shocks from permanent structural trends.
- **Unforeseeable external and extreme events:** Purely empirical machine learning models cannot anticipate unforeseen geopolitical interventions (such as regulatory price caps or export restrictions), major transmission grid outages, or unprecedented extreme weather shocks not represented in historical training data.
- **Portfolio opacity and unobserved churn:** Data reflects aggregate retailer consumption without submetering or customer segmentation (residential vs. commercial), nor customer count metadata. The model assumes a relatively stationary customer portfolio; in practice, customer acquisition or churn creates baseline demand shifts that models may misinterpret as forecast bias.
- **Meteorological parsimony:** Temperature is the sole weather variable provided. Other thermodynamic drivers affecting building heat loss and heat pump efficiency—notably wind speed (wind-chill effect), solar irradiance (passive solar gain), and relative humidity—are unobserved.
- **Deterministic weather forecast assumption:** Temperature forecasts are assumed known and error-free at nomination time, neglecting weather forecast error propagation over the 14–37 hour lead horizon.
- **Market price-taker assumption:** In simulated imbalance cost evaluations, the retailer's forecasting errors are assumed not to influence day-ahead or balancing market clearing prices.

3 Methods, analysis and technical evaluation

4 Business evaluation, value and recommendations

5 Deployment and production readiness

6 Monitoring and maintenance

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